Optimising price-based demand response to reduce carbon emissions in the UK energy market

An extended research project report submitted to the University of Manchester for the degree of MSc Data Science in the Faculty of Humanities

2025

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Abstract

Electrification represents a core strategy in the UK's progression towards net zero. However, CO_2 emissions will only be reduced if electricity is generated by increasingly clean energy sources. The non-programmable nature of renewable energy sources and the limited rollout of storage technologies pose challenges to supply-side interventions. Subsequently, demand response signifies an important tool which must be used to reduce emissions by inducing consumption flexibility. Despite this, existing research offers little exploration into how demand response can be leveraged effectively for this purpose. This study addresses this research gap by estimating marginal emission factors for the UK energy system and optimising two multiple dynamic pricing models to reduce CO_2 emissions. Simulation results from 100 days show that the developed pricing strategies yield substantial emission reductions compared to baseline levels. Moreover, when retail profits are considered as a joint objective within the optimisation formula, CO_2 emissions can be reduced significantly at no detriment to retail profitability. These findings provide evidence that price-based demand response can be an effective tool for decarbonisation and highlight its commercial viability in the energy retail market.

Acknowledgements

I would like to thank my supervisor, Fanlin Meng, for inspiration and guidance during this project. Thanks also to my close friends and family, who have offered encouragement and support throughout.

Declaration

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List of Illustrations

Figure 1: Average hourly energy generation from the UK energy system aggregated by energy source, pg15

Figure 2: Average hourly generation comparison between: years (top), seasons (middle), days (bottom), pg16

Figure 3: Exemplary regression of $\Delta\Gamma_t$ on ΔG_t considering all data points, pg19

Figure 4: Marginal emission factors and marginal generation mix vs. system load, pg26

Figure 5: Average hourly marginal emission factors and system loads, pg27

Figure 6: Histogram of changes in CO_2 emissions following two price optimisations, pg28

Figure 7: Hourly electricity prices (top) and demand profiles (bottom) when prices are optimised for the minimisation of CO_2 emissions, pg29

Figure 8: Changes in retail profits and CO_2 emissions following joint optimisation using various priority weightings, λ, pg31

Table 1: Source-specific CO_2 emission factors sourced from Rogers and Parson (2024), pg17

Table 2: Parameter settings for three different consumer groups, pg22

Table 3: Summary of optimisation results compared to the baseline, pg28

Table of Contents

1 INTRODUCTION 5

2 LITERATURE REVIEW 7

2.1 DEMAND RESPONSE 7
2.2 EMISSION FACTORS 8
2.2.1 Average emission factors 8
2.2.2 Marginal emission factors 9
2.3 REDUCING EMISSIONS WITH DEMAND RESPONSE 10
2.4 CONTRIBUTION 12

3 METHODOLOGY 14

3.1 DATA 14
3.1.1 Electricity generation by fuel type 14
3.1.2 Smart meter data 14
3.1.3 Wholesale prices 15
3.2 MARGINAL EMISSION FACTORS 15
3.2.1 Exploratory analysis 15
3.2.2 Source-specific emission factors 16
3.2.3 Marginal generation sources 18
3.2.4 Linear regression of marginal emission factors 18
3.2.5 Marginal emission factors at different loads 18
3.2.6 The marginal mix 19
3.2.7 Marginal emission factors at different times of day 19
3.3 DEMAND MODELLING 20
3.3.1 Price-demand modelling 20
3.3.2 Customer demand modelling 21
3.4 THE BASELINE 22
3.4.1 Demand 22
3.4.2 Price 23
3.4.3 Wholesale purchasing cost 23
3.5 MULTIPLE DYNAMIC PRICE OPTIMISATION 23
3.5.1 Optimisation constraints 23
3.5.2 Minimising carbon emissions 24
3.5.3 Joint optimisation 24

4 RESULTS AND ANALYSIS 25

4.1 MARGINAL EMISSION FACTORS 25
4.2 MINIMISING CARBON EMISSIONS 27
4.3 JOINT OPTIMISATION 29
4.4 DISCUSSION 31

5 CONCLUSION 33

6 REFERENCES 34

Word count: 7,207

1 Introduction

The world is facing a climate crisis. Global temperatures have increased by more than one degree above preindustrial levels and are on track to rise above three degrees by 2100 (United Nations, 2019). This current trajectory predicts an inevitable violation of the legally binding Paris Agreement of 2015.

In response, the UK has pledged to reach net zero by 2050. A commitment which requires the electrification of its energy system and full integration of renewable, non-carbon-emitting energy sources.

The National Grid estimates that in 2024, 51% of the UK electricity consumption was generated by zero-carbon energy sources (NESO, 2025). Whilst this is a significant proportion, increasing the share of low-carbon and renewable energy sources (RES) further faces challenges. RES are non-programmable. Unlike traditional fossil fuels, we cannot dictate the timing or quantity of their generation. Peaks in RES generation do not typically align with peaks in demand.

Temporal displacement of supply and demand for clean energy can be corrected by storage technologies which preserve generation until consumption. Pumped storage achieves this goal, but its expansion is bound by geographical and physical constraints. Emerging technologies such as lithium-ion batteries and green hydrogen are promising solutions but are not yet financially feasible on an industrial scale.

Demand response (DR) is recognised as a primary source of consumption flexibility. Amidst supply-side challenges, it represents an important tool to increase immediate RES penetration. DR in the energy market is widely studied, but research has predominantly focused on increasing retail profits.

Most research that considers DR's environmental potential is insufficient in one of two ways. First, emissions reductions are sought by decreasing aggregate consumption instead of flexible load shifting. Second, the impacts of consumption changes are evaluated using average emission factors which have been shown to overestimate environmental benefits in comparison to marginal emission factors.

With this motivation, the purpose of this research is to use marginal emission factors to explore how DR can reduce carbon (CO_2) emissions by increasing the flexibility of energy

consumption. Moreover, by considering the impacts on retail profitability and consumer costs, it aims to comprehensively evaluate the feasibility of climate-optimised DR in the commercial retail market.

The rest of this paper is organised as follows. Section 2 reviews the existing literature pertinent to the study's themes. Section 3 details the research methodology. Section 4 presents, analyses, and discusses the results before Section 5 offers concluding remarks.

2 Literature Review

This section presents literature regarding the central components of this paper. Section 2.1 provides an overview of DR in the energy market. Section 2.2 discusses the appropriate methods for estimating emission factors from the UK energy system. Section 2.3 presents existing research on DR designed explicitly to reduce CO_2 emissions and Section 2.4 summarises and details the contribution of this paper.

2.1 Demand Response

DR refers to the optimal allocation of consumption by consumers in response to varying incentives or prices designed by retailers to indirectly achieve a given objective (Deng et al., 2015). Through its ability to shift demand loads, DR is viewed as an effective source of consumption flexibility in the energy market (Fang et al., 2012). Amongst various aims, retailers typically use DR to flatten peaks in demand curves, allowing them to source energy at lower costs (De Caro et al., 2019).

Price-based demand response (PBDR) is the most common example of DR in the energy market. By varying prices through the day, PBDR divides the energy supply into separate products, aiming to influence consumer consumption habits (Stoll et al., 2014).

Time of Use (ToU) tariffs present a common application of PBDR, which sets different prices at fixed intervals, often corresponding to peak and off-peak hours (Nicolson et al., 2018). However, the scope of ToU is limited. The simplicity of its pricing structure restricts its ability to reduce peaks and can cause rebound peaks if prices do not adapt to changes in off-peak demand (Faruqui et al., 2010).

Due to the limitations of ToU, research is paying greater attention to dynamic pricing models (Meng et al., 2023). For example, dynamic Time of Use (dToU) removes the constraint of fixed time intervals, allowing peak and off-peak periods to vary on a daily basis (Nicolson et al., 2018). This is developed further in real time pricing (RTP) models, which continually adjust prices on a sub-hourly basis (Muratori and Rizzoni, 2016). By offering more detailed and adaptable pricing structures, dynamic pricing makes it easier for price-setters to achieve their objectives (Khan et al., 2016).

Regarding dynamic pricing, most research has focused on uniform tariffs, which offer the same prices to all customers. However, in addition to adjusting the timing and price level, energy tariffs can also be varied across different customer groups. Multiple pricing models segment the customer base and offer different tariffs to different groups. By considering the unique characteristics of different consumers, research suggests that multiple pricing can outperform uniform tariffs, by offering 'the right prices to the right consumer' (Meng et al., 2023). This study seeks to build on this research by assessing the capacity of multiple dynamic tariffs to facilitate CO_2 emission reduction.

2.2 Emission Factors

Research into PBDR is developing increasingly sophisticated pricing models to induce greater demand side flexibility in support of a price-setting objective. To inform how these strategies can achieve an environmental benefit, this section discusses the role of emission factors. Emission factors quantify the amount of emissions released by the generation or consumption of a unit of energy [g/kWh] (Zheng et al., 2015). Their accurate estimation is essential to quantifying the environmental effect of electric load shifting in response to PBDR intervention (Ryan et al., 2016).

2.2.1 Average emission factors

The impact of environmental policy is commonly estimated using average emission factors (XEFs), which describe the ratio of emissions to energy generated for a given period (Broekhoff, 2007). Using XEFs implicitly assumes that all emissions from the electricity grid over a given time frame are proportionally shared by all electricity consumers. (Baumgartner, 2019). This makes them a simple and useful tool for calculating overall emission balances of systems or consumers (Fleschutz et al. 2021).

However, XEFs are unreliable when estimating the reduction in emissions created by demand-side intervention (Hawkes, 2010; Siler-Evans et al., 2012). Using XEFs to calculate the effect of consumption changes makes an invalid assumption that changes in demand affect all elements of the energy system equally (Hawkes, 2010). This ignores the merit-order, which dispatches power plants in the order of lowest marginal generation cost. In reality, only the cheapest available (marginal) generators react to load changes (Schram et al., 2020). By attributing an even distribution of emissions across all generation units, using XEFs neglects

differences between fuel-specific emission factors, therefore misestimating the effect of a change in load (Roselli et al., 2020).

2.2.2 Marginal emission factors

In contrast, marginal emission factors (MEFs) describe how emissions change with respect to a change in system load (Zivin et al., 2014). MEFs can more accurately account for changes in consumption load because they acknowledge the merit order of generation systems. By identifying marginal generators for a given load, they can determine the amount of emissions avoided or created by a decrease or increase in consumption (Siler-Evans et al., 2012).

The accurate evaluation of emission reduction strategies relies on the accuracy of the MEFs estimates used (Hawkes, 2010). Two methods for estimating MEFs have emerged in the literature (Zheng et al. 2015).

Voorspools and D'Haeseleer (2000) and Marnay et al. (2002) were amongst the first studies to produce MEF estimates. They adopted the 'merit order approach' which identifies the marginal generators which respond to demand changes. MEFs are then equated to the specific emission factor of the most recently dispatched generation unit (Huber et al., 2021).

Rekkas (2005) introduced a method based solely on historical data on the dispatch of generators aggregated by fuel and technology types in Great Britain. He used a linear regression model, to estimate the changes in each energy source to a change in system load. Once fuel-specific behaviour was understood, fuel-specific emission factors were used to obtain MEFs.

Application of the merit order approach can be challenging because it demands a detailed understanding of complex energy systems. Doucette and McCulloch (2011) highlight the difficulty of empirically identifying the marginal power plant. In contrast, the regression approach requires less detailed knowledge of energy system characteristics because it implicitly models complexities such as trading decisions, logistical constraints and transmission and distribution restrictions (Hawkes, 2010; Siler-Evans et al., 2012).

A further drawback of the merit order method is the assumption that emission factors follow a discontinuous step function. Aggregate system load changes are met by combinations of generators. Therefore, changes in emissions likely follow a smoother gradient (Hawkes, 2010).

Following popularisation by Hawkes (2010), regression-based MEF modelling has been extended. Siler-Evans et al. (2012) estimate a more comprehensive emission factor by

incorporating nitrogen oxide and sulphur dioxide alongside CO_2. Zheng et al. (2015) model the ramp rate constraint of bulk generators for more detailed short-term responses. Péan et al. (2018) fit a quadratic model in contrast to most linear designs.

In some cases, the merit order approach is more appropriate than regression models. For long-term MEF forecasts, merit orders models can account for the structural changes determined by the emissions factors such as downscaling of power plants, which is unobserved by regressions (Thind et al., 2017). In addition, the need for a detailed database of energy generation aggregated by fuel type makes regression approaches unsuitable for many regions (Siler-Evans et al., 2012).

Overall, regression-based approaches allow for more complex modelling (Rekkas, 2005) and are easier to compute than merit order alternatives (Yang, 2013). Given the availability of detailed UK generation data aggregated by fuel type and the short-term focus of DR intervention, regression-based MEF estimation is best suited to this study.

2.3 Reducing Emissions with Demand Response

Building upon the previous discussion, this section analyses the existing research which considers DR strategies optimised to reduce CO_2 emissions. Evaluation of DR's environmental potential is mostly restricted to the effects of decreased energy consumption (Vardakas et al., 2014). Although this achieves the goal of reducing emissions, it represents a suboptimal solution compared to an equivalent reduction achieved load shifting.

Fouladfar et al. (2019) consider the impact of DR on reducing emissions from a residential neighbourhood microgrid. However, like many studies, effects are calculated using XEFs opposed to MEFs, reducing the reliability of results.

Multiple studies propose and develop incentive-based mechanisms to reduce emissions from consumption. Leerbeck et al. (2020) develop 24-hour ahead MEF forecasts designed for consumers to optimise scheduling of flexible applications such as heat pumps. Similarly, Huber et al. (2021) measure the benefits of MEF signalling for smart charging of electric vehicles in Germany. Both papers conclude that providing direct MEF signals is a powerful tool to reduce CO_2 emissions by incentivising load shifting.

The significance of these studies relies on consumers being motivated to reduce their carbon footprint and being willing to consume flexibly. Whilst consumption flexibility increases when incentivised by greater RES integration (Huber et al., 2019) MEF signals often encourage load

shifts to times of higher energy prices (Huber et al., 2021). When faced with increased energy bills, the willingness of consumers to respond to environmental incentives becomes more doubtful.

Regarding price-based incentives, Broeer et al. (2018) run agent-based simulations to test how PBDR can be used to increase integration from wind generation. They conclude a small benefit to variable RES penetration but also note the potential for a smaller adverse effect. Fleschutz et al. (2021) model MEFs from twenty energy systems across Europe and run simulations to show that PBDR would increase CO_2 emissions for eight. These results reflect a modest potential for PBDR to reduce CO_2 emissions.

Mirroring findings from Huber et al. (2021), Fleschutz et al. (2021) conclude that PBDR will increase emissions in countries where there is a negative relationship between MEFs and energy prices. This describes the merit order dilemma, the scenario that marginal fuel sources with higher emission factors, are prioritised in the merit order due to lower generation costs (Regett et al., 2018). Existence of the dilemma has prompted the argument that PBDR cannot achieve emission reduction (Regett et al., 2018; Fleschutz et al. 2021).

However, both Broeer et al. (2018) and Fleschutz et al. (2021) model energy prices exogenously based off wholesale market costs. In this case, when the merit order dilemma prevails, emissions and cost minimisation become incompatible objectives (Leerbeck et al., 2020). Neither study proposes an active pricing strategy to specifically improve emission reduction. To assess the full potential of PBDR, pricing strategies must be optimised so prices can incentivise consumption from cleaner energy sources. Since optimised prices are not bound to generation costs, they can be aligned with lower MEFs and be used to reduce CO_2 emissions even in the presence of the merit order dilemma.

After an extensive search, Pepiciello et al. (2019) was the only study found which develops a pricing strategy optimised for emission reduction. Through a year-long simulation they found that implementing a real time pricing (RTP) model driven by MEFs substantially reduced CO_2 emissions compared to the baseline ToU tariff. By linking prices to MEF estimates, this result suggests, in contrast to Broeer et al. (2018) and Fleschutz et al. (2021), that PBDR has significant potential as an emission reduction strategy.

2.4Contribution

Overall, limited research has considered the merits of DR for the reduction of CO_2 emissions from the energy system. Current evidence suggests that PBDR policies have the greatest potential, but Pepiciello et al. (2019) present the only research which optimises a pricing strategy based on MEF estimates.

An important consideration for all PBDR policies is the economic feasibility of the pricing model. However, existing research into PBDR as a tool for emissions reduction pays limited attention to the role of the retailer. Energy retailers set prices to maximise profit, therefore MEF-driven prices are unlikely to be implemented if they decrease profitability, in spite of environmental benefits. Whilst Pepiciello et al. (2019) produce optimistic results, without assessing the economic feasibility of their pricing strategy, the significance of their findings is uncertain.

Based on the scope of existing research, this study aims to contribute to the literature by answering four specific research questions. First, accurate time-varying MEF estimations are required to develop and evaluate the environmental impact of PBDR models. Understanding the behaviour and mechanisms behind MEF estimates helps to evaluate the outcomes of MEF-driven pricing models. Therefore, this paper begins by asking:

> RQ1: What are the characteristics of marginal emission factors in the UK energy system?

Given that only one previous study has produced an energy pricing model to explicitly reduce CO_2 emissions, its findings are tested via a similar method. By optimising a multiple dynamic pricing model, this paper answers:

> RQ2: To what extent can price-based demand response reduce CO_2 emissions?

The literature review highlights that existing research fails to consider the economic feasibility of the PBDR strategies developed. Therefore, this study introduces a novel pricing strategy developed via the joint-optimisation of minimising CO_2 emissions whilst maximising retail profits. In doing so it answers the question:

> RQ3: Is it feasible to reduce CO_2 emissions through price-based demand response in the energy market?

Finally, to complement the previous question, this paper introduces a discussion on the relationship between retail profits and CO_2 emissions with respect to price-setting. Understanding how PBDR can balance both objectives is necessary to consider the full range of available pricing strategies. Therefore, the final question asks:

> RQ4: What are the trade-offs between CO_2 emissions and retail profits relating to price-based demand response in the energy market?

3 Methodology

This section describes the methodology used to answer the identified research questions. Section 3.1 provides a description of the data. Section 3.2 outlines regression-based MEF estimation. Section 3.3 discusses customer demand modelling and Section 3.4 calculates baseline results. Last, section 3.5 describes the optimisation of two multiple dynamic pricing models.

3.1 Data

This study relies on three publicly available datasets. A fuel-specific dataset of historical electricity generation supports regression-based MEF estimation. A smart meter dataset is used to sample representative daily demand loads and market index price data is used as an index for wholesale electricity prices.

3.1.1 Electricity generation by fuel type

Historic electricity generation data is sourced from Elexon (2025a) through their Balancing Mechanism Reporting Service, which records fuel-specific power outputs [MW] averaged over 30-minute intervals. Due to variation between the emission factors of electricity generation from importing countries (Bruce et al., 2024), the detailed aggregation of interconnectors makes this data set the best source currently available for estimating MEFs.

The chosen dataset underestimates contributions from renewable sources because it excludes generation embedded into the grid (Elexon, 2025a). Subsequently, it omits solar and a significant proportion of wind generation. However, because embedded sources are prioritised in the merit-order, their exclusion does not affect MEF calculations.

Data from this dataset is selected for the three years spanning between 1st August 2022 – 31 July 2025. This time frame is chosen to provide a sufficient number of data points whilst offering an accurate reflection of the current energy system. Selecting complete 12-month cycles ensured seasonal effects are equally balanced.

3.1.2 Smart meter data

Household smart meter data is sourced from UK Power Network's (2014) Low Carbon London project. Observations include residential energy consumption readings [kWh] at half-hourly intervals from 5,567 London households. The dataset indicates whether customers were facing

a flat rate or a dToU energy tariff at the time of each reading. This is the best currently available dataset containing residential consumption data under price variation.

3.1.3 Wholesale prices

Wholesale, short-term energy prices [£/MWh] are taken from Elexon's (2025b) market index prices (MIP) dataset. MIPs represent the average price of electricity trades weighted by volume during a given settlement period. Observations are recorded every 30 minutes and exclude long-term forwards and do not account for bilateral trades. MIP data is selected from 24/07/25 - 31/07/25 which corresponds to the most recent week within the time frame of the energy generation dataset.

3.2 Marginal Emission Factors

This section estimates MEFs for the UK energy system, following the linear regression approach pioneered by Rekkas (2005) and Hawkes (2010).

3.2.1 Exploratory analysis

The stacked area chart in Figure 1 plots the average hourly electricity generation [GW] for the UK energy system. It illustrates the role of gas as the largest generation source and as the greatest contributor to aggregate load variation. Wind and nuclear production also represent prominent sources, but their contributions are typically consistent and unrelated to demand on the system. Following gas, pumped storage has the highest proportional variance. This is reflected by notable contributions on the crests of the morning and evening peaks.

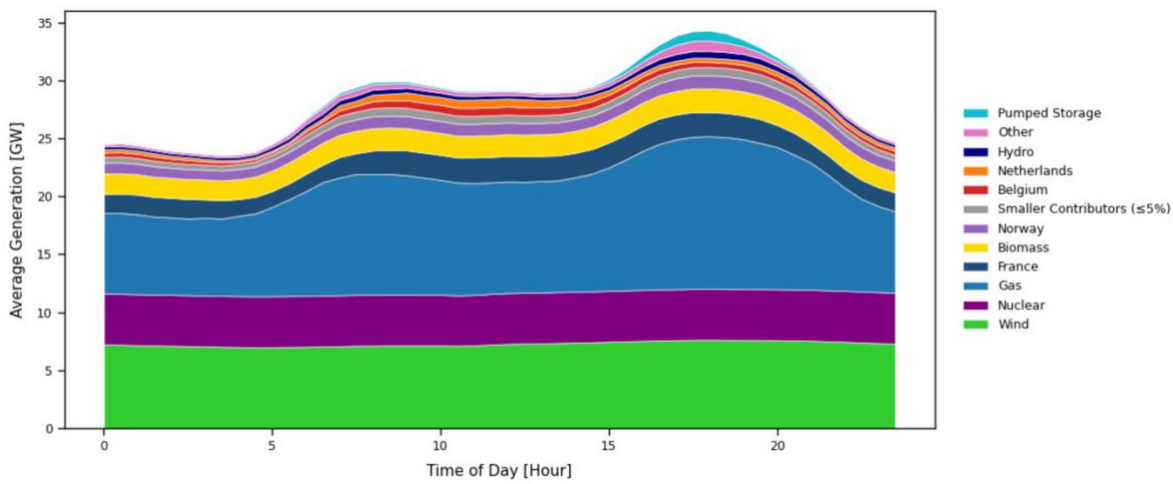

Figure 1: Average hourly energy generation from the UK energy system aggregated by energy source

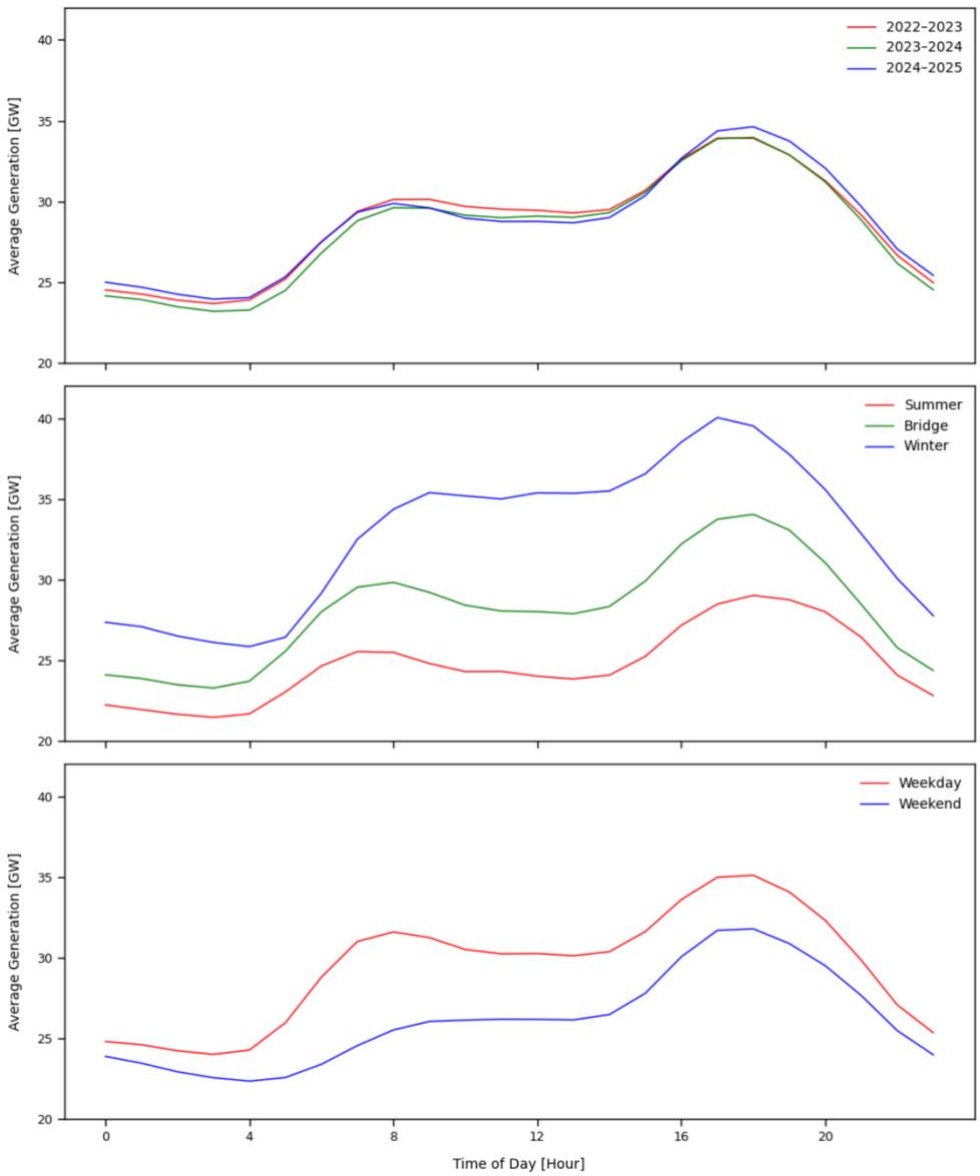

Figure 2: Average hourly generation comparison between: years (top), seasons (middle), days (bottom).

Temporal variation in energy generation also prevails at broader resolutions. Figure 2 presents hourly generation curves partitioned by years (top), seasons (middle) and days (bottom). The plots show little variation across years, but significant differences in the size and shape of generation curves when seasonality and weekends are controlled for. Intuitively, winters demand greater energy consumption than summers, and bridge months fall in between. Compared to weekdays, less energy is generated over weekends, and the load curve does not exhibit a morning peak.

3.2.2 Source-specific emission factors

Source-specific CO_2 emission factors measure the rate at which carbon dioxide is emitted during electricity generation from a given source k. The Elexon dataset does not observe the

CO_2 emissions produced as a byproduct of energy generation therefore this study relies on emission factors estimated by Rogers and Parson (2024). These estimations (Table 1) are used to calculate carbon emissions and carbon intensity variables. Energy sources are divided between domestic generation, comprised of specific fuel types, and foreign generation, comprised of nations whose exports have varying emission factors determined by the characteristics of their domestic generation mix.

Table 1: Source-specific CO_2 emission factors sourced from Rogers and Parson (2024).

Fuel Type	γ^k [gCO_2/kWh]	Energy Importer	γ^k [gCO_2/kWh]
Biomass	120	Belgium	167
Coal	937	Denmark	180
Gas	394	France	85
Hydro	0	Ireland	346
Nuclear	0	Netherlands	354
Other	300	Northern Ireland	346
Pumped storage	0	Norway	29
Wind	0		

Total carbon emissions Γ_t [gCO_2] at the period t depend on source-specific emission factors γ^k [gCO_2/kWh] and electricity generation G_t^k [gCO_2]:

$$\Gamma_t = \sum_k \left(\gamma^k \cdot G_t^k\right). \tag{1}$$

The average carbon intensity AEF_t [gCO_2/kWh] at period t relates to total carbon emissions Γ_t [gCO_2] and the total electricity generation G_t [kWh]:

$$\mathrm{AEF}_t = \frac{\Gamma_t}{G_t} \tag{2}$$

where:

$$G_t = \sum_k G_t^k. \tag{3}$$

3.2.3 Marginal generation sources

When calculating MEFs, regression models should only consider generation from sources which adapt their volume of generation in response to changes in demand from the grid. Marginal energy generation excludes sources with the lowest cost of production because they are prioritised in the merit order. Therefore, in keeping with Hawkes (2010) and McCarthy and Yang (2010) wind and nuclear generation are excluded from the mix of energy generation sources.

3.2.4 Linear regression of marginal emission factors

The conventional linear regression describes how CO_2 emissions change in response to a change in energy generation over a given time period:

$$\Delta\Gamma_t = \beta \cdot \Delta G_t + \epsilon_t \tag{4}$$

where:

$$\Delta\Gamma_t = \Gamma_t - \Gamma_{t-1} \tag{5}$$

and:

$$\Delta G_t = G_t - G_{t-1}. \tag{6}$$

This methodology is exemplified by Figure 3, which includes all observations from the dataset. β is interpreted as the average MEF of the energy system and visualised by the slope of the regression model.

3.2.5 Marginal emission factors at different loads

As the load on the energy system increases, generation works up the merit order, reaching energy sources with increasingly high costs of production. The mix of energy sources used to generate the marginal kWh of energy changes, creating different MEFs for different levels of system load.

Subsequently, observations are partitioned into 2.5% quantile bins b, based on the total generation load from all marginal generation sources. Separate regression models are run for each quantile, yielding 40 estimates corresponding to the average MEF for a specific generation load range:

$$\Delta\Gamma_b = \beta_b \cdot \Delta G_b + \epsilon_b. \tag{7}$$

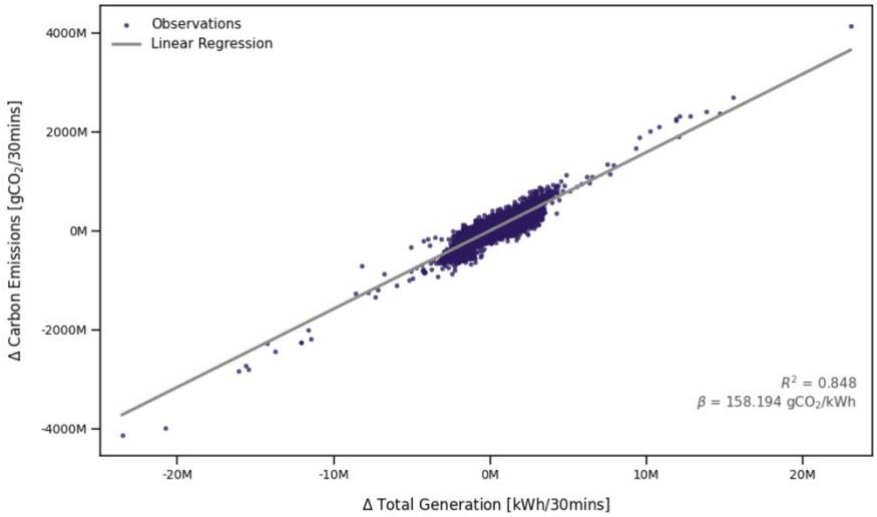

Figure 3: Exemplary regression of $\Delta\Gamma_t$ on ΔG_t considering all data points. Style adopted from Huber et al. (2020).

3.2.6 The marginal mix

Precise definitions and calculations of the marginal mix varies within the literature (Huber et al., 2021). For this paper the marginal mix refers to the collection of fuel sources which generate the marginal kWh of energy in their relative proportions. Therefore, it is intrinsically linked to MEF estimation. The marginal mix explains how electricity is generated providing a clearer understanding of how and why MEFs change.

Calculating the marginal mix requires consideration of source-specific generation changes relative to total generation. Here it was calculated using the pre-defined generation load bins:

$$\Delta G_b^k = \beta_b^k \cdot \Delta G_b + \epsilon_b^k \tag{8}$$

Marginal contributions β_b^k are normalised and are presented as percentages in Figure 4.

3.2.7 Marginal emission factors at different times of day

Due to the significant variation between weekday and weekend generation (Figure 2), weekends are not considered when evaluating MEF at different times of day. The remaining observations within the energy generation dataset are assigned an MEF estimate corresponding to the predefined 2.5% load quantiles. This creates an MEF time series for which the mean hourly estimates are presented in Figure 5.

3.3 Demand Modelling

Price optimisation requires demand functions to model how price-sensitive consumers adjust their consumption behaviour in response to changes in their energy tariff. This section details the methods followed to produce the demand functions embedded into the optimisation models.

3.3.1 Price-demand modelling

The methodology adopts a straight-forward linear demand model which is favourable within the literature (Meng et al., 2023). Though simplistic, it is easily implemented, has strong interpretability, and contributes to well-defined optimisation problems. Subsequently, the consumption demand of consumers g during hour h when faced with a set of dynamic prices $P^g = \{p_1^g, p_2^g, \dots, p_H^g\}$ is defined as:

$$q_h^g(p_1^g, p_2^g, \dots, p_H^g) = \alpha_h^g + \varepsilon_{h1}^g p_1^g + \cdots + \varepsilon_{hh}^g p_h^g + \cdots + \varepsilon_{hH}^g p_H^g. \tag{9}$$

α_h^g represents a constant intercept term. ε_{hl}^g is a price elasticity which determines the relationship between the price at hour l and quantity demanded at hour h:

$$\varepsilon_{hl}^g = \frac{\Delta q_h^g}{\Delta p_l^g}. \tag{10}$$

Where $l = h$, ε_{hl}^g represents a self-elasticity. Otherwise ($l \neq h$), ε_{hl}^g measures cross-prices elasticities. As an extension, changes in demand can be expressed as:

$$\Delta q_h^g = \sum_{l=1}^{H} \varepsilon_{hl}^g \, \Delta p_l^g. \tag{11}$$

All elasticities are defined in absolute terms. That is, for a cross-price elasticity (ε_{hl}^g) of 0.1, a price increase of 1p/kWh at hour l will increase demand at hour h by 0.1kWh, ceteris paribus.

If we assume that demand shifts in relation to price changes are confined within one 24-hour cycle, an individual's daily demand can be expressed by the self-contained function:

$$\Delta Q^g = \mathcal{E}^g \Delta P^g \tag{12}$$

where Q^g and P^g are 24-dimensional column vectors of hourly demand quantities and hourly prices, respectively. $\mathcal{E}^g$ is a 24×24 matrix comprised of row vectors ε_h^g which contain price elasticities relative to h.

3.3.2 Customer demand modelling

Due to a lack of consumption data under-price variability, the smart meter dataset cannot be used to directly produce household-level demand functions. Instead, exogenous elasticity matrices are created to model customers with different price-sensitive behaviours. In a similar method to Meng et al. (2023), three distinct consumer profiles are defined.

3.3.2.1 Constraints

Before elasticity matrices are created, a series of constraints are imposed to ensure customer demand models illicit rational market behaviours. Self-elasticities (diagonal elements) are made non-positive (13) and cross-price elasticities are made non-negative (14):

$$\varepsilon_{hh}^{g} \leq 0. \tag{13}$$

$$\varepsilon_{hl}^{g} \geq 0 \text{ if } h \neq l. \tag{14}$$

(13) defines hourly demand as an ordinary good. When the price increases in hour h and other prices remain unchanged, demand at h will not increase. Following this price change, a consumer might shift consumption from hour h to hour l, but is unlikely to decrease consumption at hour l. This is reflected by constraint (14).

In addition, for all elasticity matrices, the sum of the elements across each column must equal zero:

$$\sum_{h=1}^{24} \varepsilon_{hl}^{g} = 0 \quad \forall l. \tag{15}$$

Introducing price changes might increase or decrease aggregate demand depending on how they align with consumption preferences. To isolate the impact of prices on CO_2 emissions clearly, constraint (15) ensures demand conservation. Price changes can promote consumption reallocation but cannot change consumption volume over 24 hours i.e. the system is lossless.

3.3.2.2 Elasticity modelling

Three different elasticity matrices are modelled to represent the characteristics of three consumer groups. These characteristics are defined by the parameters ϕ^{g} and σ^{g}. ϕ^{g} represents the magnitude of non-positive self-elasticities whereby larger values reflect greater responsiveness to price changes.

σ^g is a flexibility parameter which defines the variance of a Gaussian kernel used to determine the values of cross-price elasticities. Small σ^g values create a narrow shifting window whilst larger values reflect consumers who have the flexibility to shift electricity over a wider period.

Mirroring Meng et al. (2023), this paper considers a residential district of 3,000 consumers with 1,000 from each of three consumer groups, with each group exhibiting varying degrees of price elasticity and consumption flexibility. Insensitive consumers (IS) are typically inelastic to price changes and have limited flexibility. Sensitive consumers (SC) are more elastic to price changes than IS consumers and have a broader shifting window. Highly sensitive consumers (HS) are highly elastic and have the widest shifting window. These characteristics are coded by the parameter choices presented in Table 2.

Table 2: Parameter settings for three different consumer groups

Consumer group, g	ϕ^g	σ^g
Insensitive (IS)	0.002	1.5
Sensitive (SC)	0.02	2
Highly sensitive (HS)	0.06	4

3.4 The Baseline

The demand function expressed in equation (12) calculates a change in consumption following a change in price. To yield meaningful results, changes are contextualised according to initial (baseline) levels of demand and price. This section details the conditions for the baseline results presented in Table 3.

3.4.1 Demand

From the smart meter dataset only observations corresponding to the flat rate tariff (14.223pkWh) were considered when finding representative (baseline) daily demand profiles. This allows for the reasonable assumption that, in the absence of price variation, the data reflects consumers' optimal allocation of daily energy consumption.

The remaining data points were processed into 24-dimensional vectors $q^i = \{q_1^i, q_2^i, \dots, q_H^i\}$ with elements reflecting energy consumption from the individual household i at hourly intervals. 1,000 households are randomly sampled, and 100 daily (weekday) demand profiles are randomly drawn for each household. Demand profiles from each day d are aggregated at an hourly resolution to create 100 baseline daily aggregate demand vectors $Q^d =$

$\{Q_1^d, Q_2^d, \dots, Q_H^d\}$ reflecting the total energy demand across all 1,000 households. To prevent variation between consumer groups, the same aggregate demand was used for all of them.

3.4.2 Price

The initial (baseline) set of daily energy prices $P^0 = \{p_1^0, p_2^0, \dots, p_H^0\}$ faced by each consumer group is a flat rate tariff to mirror the conditions corresponding to the representative demand profiles. The baseline tariff P^0 is set to 15p/kWh.

3.4.3 Wholesale purchasing cost

Market index price data consists of vectors $c^d = \{c_1^d, c_2^d, \dots, c_H^d\}$ containing hourly wholesale costs. Representative purchasing costs were defined as the mean vector $\bar{c} = \{\bar{c}_1, \bar{c}_2, \dots, \bar{c}_H\}$ across weekday prices and are used within all profit calculations.

3.5 Multiple Dynamic Price Optimisation

Using the customer demand functions modelled in Section 3.3.2, price optimisation models are run to answer RQ2-4. Hourly prices are optimised independently for each group using off-the-shelf optimisation packages. Results are aggregated across groups to calculate changes in emissions, profit, and consumer energy bills. Each optimisation is simulated 100 times for randomly sampled vectors of daily MEF estimates, with the mean outcomes presented in Table 3.

3.5.1 Optimisation constraints

All optimisation models are subject to two price constraints. The average hourly price must remain equivalent to the baseline, flat rate tariff P^0 (15p/kWh). This ensures that positive results cannot be manufactured by simply raising aggregate prices. Moreover, it imposes fairness by forcing the retailer to offer all customers the same average price:

$$\frac{\sum_{h \in H} p_h^i}{H} = P^0. \tag{16}$$

The minimum bound is dictated by the purchasing cost $\bar{c}_h$, to prevent loss-making. The maximum bound p_h^{max} is set to 25p/kWh. This is guided by the UK energy price which stood at 25.73p/kWh at the time of writing (Ofgem, 2025).

$$\bar{c}_h \leq p_h^h \leq p_h^{max} \quad \forall g \in G, \forall h \in H, \tag{17}$$

3.5.2 Minimising carbon emissions

Subject to constraints (16) and (17), a change in CO_2 emissions for each consumer group is minimised by solving the linear programming problem:

$$\min \; MEF^T \mathcal{E}^g \Delta P^g \,, \tag{18}$$

where MEF is a vector of marginal emission factors at h estimate in Section 3.2.7.

3.5.3 Joint optimisation

Under the same constraints, an increase in retailer profits from each consumer group g can be maximised by solving the quadratic programming problem:

$$\max \; q^0 \Delta P^{g^{\mathrm{T}}} + \mathcal{E}^g \Delta P^g. \tag{19}$$

Subsequently, both of the aforementioned models can be embedded within a weighted-sum dual-optimisation problem:

$$\max \quad \lambda \left(q^0 \Delta p^{g^{\mathrm{T}}} + \mathcal{E}^g \Delta p^g) - (1-\lambda)(MEF^T \mathcal{E}^g \Delta p^g \right), \tag{20}$$

which is to find prices which simultaneously increase retailer profits whilst reducing CO_2 emission from energy consumption.

The weighted joint optimisation model is run for multiple λ values in the range [0,1] with the average results across all simulations visualised in Figure 8.

4 Results and Analysis

This section presents, analyses, and discusses the results of the methodology, providing answers to the study's research questions.

4.1 Marginal Emission Factors

Figures 4 and 5 illustrate the key outcomes from the regression-based estimation of MEFs in the UK energy system. Analysis of these results answer the first research question:

RQ1: What are the characteristics of marginal emission factors in the UK energy system?

The upper plot within Figure 4 plots the change in MEFs for the UK energy system according to a change in total load. Across 40 individual regression models, the average MEF per load quantile is 314gCO_2/kWh. The highest MEF estimate was 335gCO_2/kWh corresponding to an average system load of 32GWh. The lowest estimate was 236gCO_2/kWh, observed at an average system load of 43GWh.

The most notable trend is the steep drop in MEFs as the system load rises past 35GWh. This suggests that, as consumption levels move towards peak levels of demand, generation begins to incorporate energy sources with lower emission factors, at an increasing rate. This is a clear example of the merit order dilemma, as energy generation from cheaper sources is prioritised over cleaner, but more expensive fuel types (Regett et al., 2018).

The trend of decreasing MEFs is explained by the lower plot of Figure 4, which illustrates how the marginal generation mix changes as increasing demand is placed on the energy system. Specifically, it shows the proportional contribution of each energy source to the marginal kWh of energy generation dependent on the system load.

At most system loads, marginal generation is dominated by gas. However, the contribution from gas toward the marginal mix decreases with load as the system turns to new energy sources to meet additional demand. In particular, Figure 4 shows the increasing generation from energy sources with lower emission factors. For example, pumped storage and 'other' fuel types have fuel-specific emission factors of 0gCO_2/kWh and 300gCO_2/kWh respectively, compared to 394gCO_2/kWh for gas. Therefore, the fall in MEFs at levels of high

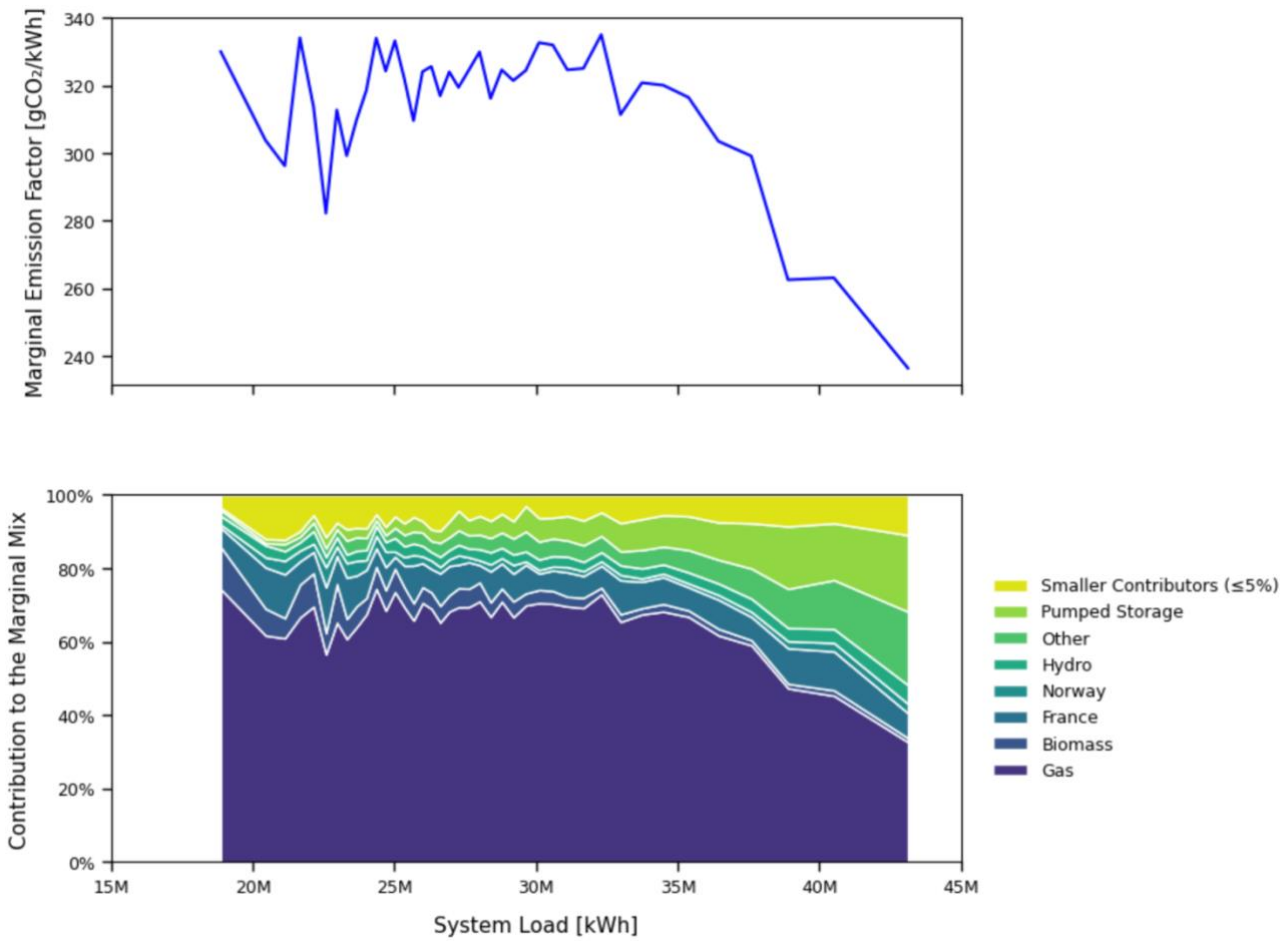

Figure 4: Marginal emission factors and marginal generation mix vs. system load. Style adopted from Thind et al. (2017).

system load is best explained by an increasing reliance on less CO_2emitting energy sources to meet marginal demand.

Based on the methodology described in Section 3.2.7, Figure 5 presents the average hourly MEF estimates in comparison to average system load. Both MEFs and consumption demand vary significantly throughout the day, eliciting a strong inverse relationship. On average, MEFs are highest between 10pm and 7am, corresponding to lower levels of system load. During peak demand between 5pm and 8pm, MEFs drop significantly, to an average below 300gCO_2/kWh.

The daily distribution of MEFs provides insight into how optimised pricing strategies might incentivise load shifting to minimise CO_2 emissions. Contrary to most PBDR implementation, it suggests that decreasing off-peak demand and increasing peak demand could be optimal. Given that hours of peak demand are associated with higher energy prices, these results highlight the requirement of MEF-driven pricing strategies for PBDR to achieve environmental objectives.

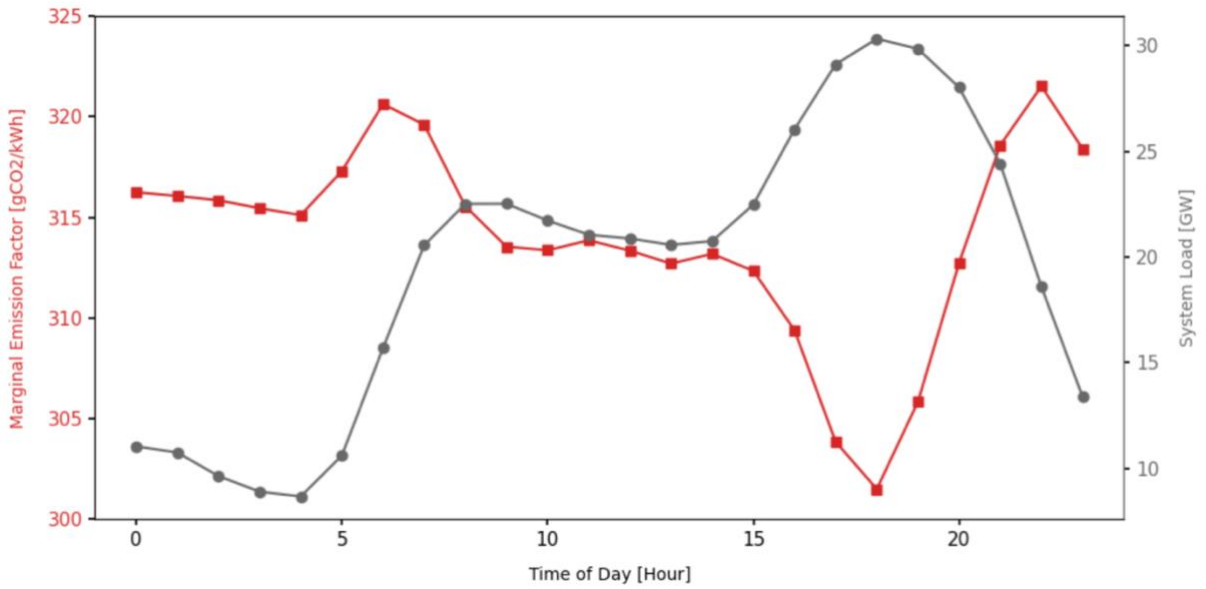

Figure 5: Average hourly marginal emission factors and system loads.

4.2 Minimising Carbon Emissions

The first pricing model described in the methodology is a multiple dynamic pricing tariff designed to minimise CO_2 emissions. This subsection compares the results from this model to the baseline, and in doing so answers the second research question:

RQ2: To what extent can price-based demand response reduce CO_2 emissions?

Table 3 presents the mean outcomes of multiple pricing optimisation from the 100 simulations. When price-setting is optimised for the minimisation of CO_2 emissions, PBDR achieves a reduction of 144kg CO_2 in comparison to the baseline, on average. This equates to a reduction 48gCO_2 per household within the modelled residential district. Emission savings are associated with an expected fall in retail profits of almost 50%. Average energy costs decrease for all consumer groups with HS consumers making the most savings compared to the baseline tariff.

The distribution of CO_2 emission savings across the 100 simulations is presented in Figure 6. The model's worst-case scenario achieved a reduction of 44kgCO_2 whilst the top 5% of cases achieved savings in excess of 353kg CO_2.

The results from this optimisation show that PBDR has the potential to reduce CO_2 emissions to a significant extent when pricing strategies are purposefully designed to incentivise consumption at times of low MEFs. This finding corroborates conclusions drawn by Pepiciello

Table 3: Summary of optimisation results compared to the baseline

Pricing model	Retailer [£]			Emissions [$kgCO_2$]	Customer bills [£]			
	Revenue	Profit	Cost	Δ	IS	SC	HS	Total
Baseline (flat rate)	4,340	1,908	2,432	(-) 0	1,447	1,447	1,447	4,340
Emission minimisation	3,386	966	2,420	(-) 144	1,424	1390	823	3,386
Joint optimisation ($\lambda = 0.6191$)	4,320	1909	2,412	(-) 98	1,609	1422	1290	4,320

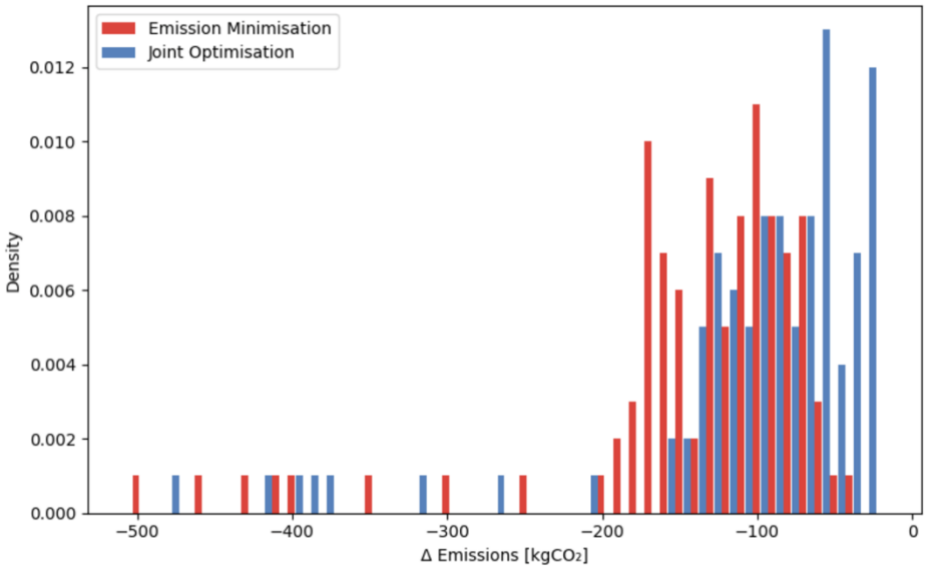

Figure 6: Histogram of changes in CO_2 emissions following two price optimisations

et al. (2019) and challenges the assertion that PBDR is an inadequate tool for emission reduction strategy (Fleschutz et al., 2021).

To provide a mechanistic understanding of how this pricing model facilitates reduced CO_2 emissions, Figure 7 presents an example of optimised prices (top) and subsequent demand loads (bottom), for each consumer group from a randomly drawn simulation. As expected, during peak demand in the evening, prices are reduced to incentive consumption when MEFs are at their lowest. This is reflected by increases in demand during these hours, creating accentuated consumption peaks. The difference in demand curves across consumer groups clearly demonstrates variations in price sensitivity. IS consumers make small adjustments to the new prices whilst SC and HS consumer show greater responsiveness.

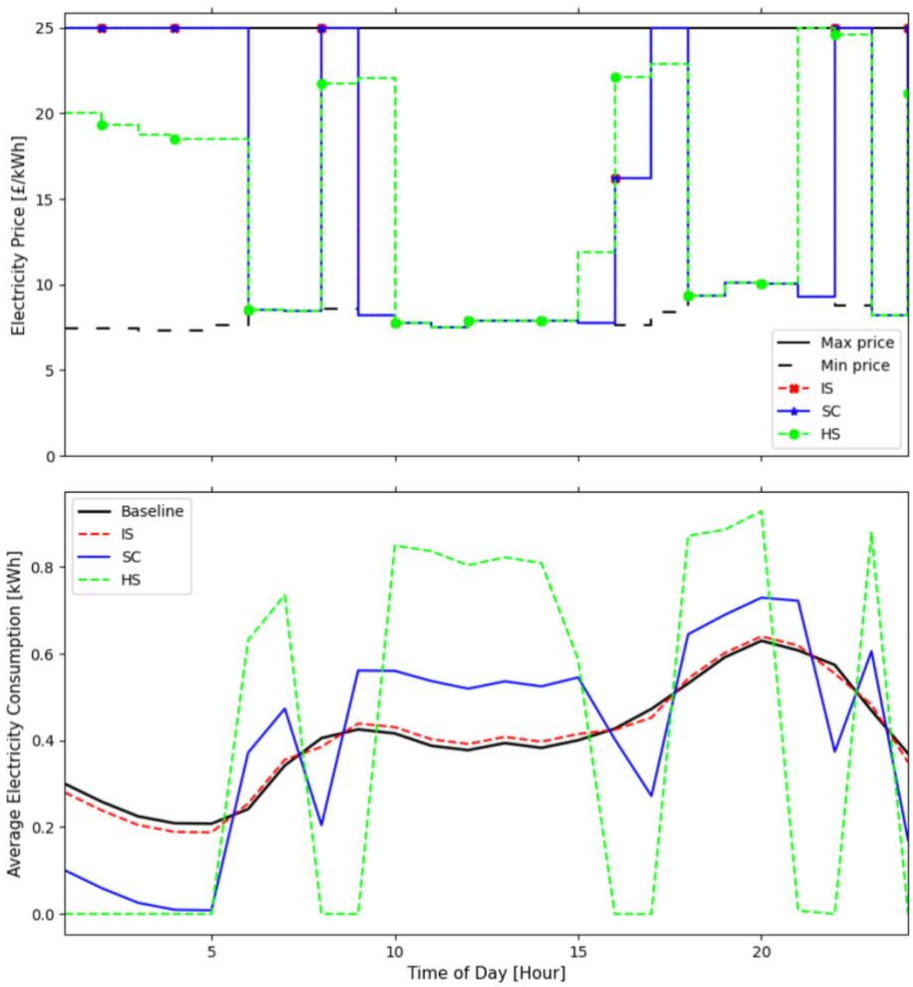

Figure 7: Hourly electricity prices (top) and demand profiles (bottom) when prices are optimised for the minimisation of CO_2 emissions.

4.3 Joint Optimisation

Following from the previous minimisation of CO_2 emissions, this section reviews the results from the joint optimisation pricing model, which aims to balance the objectives of emission minimisation and profit maximisation. In doing so it answers the remaining research questions, starting with:

RQ3: Is it feasible to reduce CO_2 emissions through price-based demand response in the energy market?

The joint optimisation pricing model produced a range of results corresponding to 21 λ values in the range $[0,1]$. The results printed in Table 3, and the distribution in Figure 6, reflect outcomes at $\lambda = 0.6191$ - the value at which retail profits are unchanged relative to the baseline.

Reading from Table 3, optimisation of the dual-objective pricing model lowered CO_2 emissions by an average of 98kg CO_2 compared to the baseline, whilst retaining an equivalent retail profit. CO_2 emissions were reduced across all simulations at $\lambda = 0.6191$, and the best 5% of cases reduced emissions by over 318kg CO_2.

Alongside profitability, an assessment of commercial viability also requires consideration of consumer costs. On average, when profits are held at baseline levels, daily energy bills rise by 11% for IS customers but decrease for SC and HS customers. Overall, total consumer costs marginally decrease by less than 1%.

The results from the joint optimisation pricing model illustrate that PBDR for the reduction of CO_2 emissions can feasibly be implemented by retailers in the energy market. When pricing strategies are driven solely by MEFs they risk sacrificing retail profit. Such models have limited value because they are unlikely to be adopted by retailers in practice. By including retail profits within pricing optimisation formula, this study introduces a commercially viable PBDR strategy which also offers a substantial environmental benefit.

Discussion to this point has focused on results from the profit maintaining optimisation at $\lambda = 0.6191$. Evaluating the broader range of joint optimisation results provides a more nuanced evaluation in response to the final question:

RQ4: What are the trade-offs between CO_2 emissions and retail profits relating to price-based demand response in the energy market?

Figure 8 plots the average outcomes of the joint optimisation over the range of λ values. Recall that when $\lambda = 1$, the pricing model disregards CO_2 emissions and prices are set at profit maximising values. Likewise, when $\lambda = 0$, retail profits are disregarded, and optimisation produces identical results to those discussed in Section 4.2.

The curve connecting both endpoints represents the trade-off between changes in retail profits and changes in CO_2 emissions. This is illustrated as smooth, positive, non-linear relationship. When optimisation places greater emphasis on profitability (higher λ) CO_2 emissions are forced to rise. Likewise, when emission reduction becomes an increased priority, retail profits are forced to decrease.

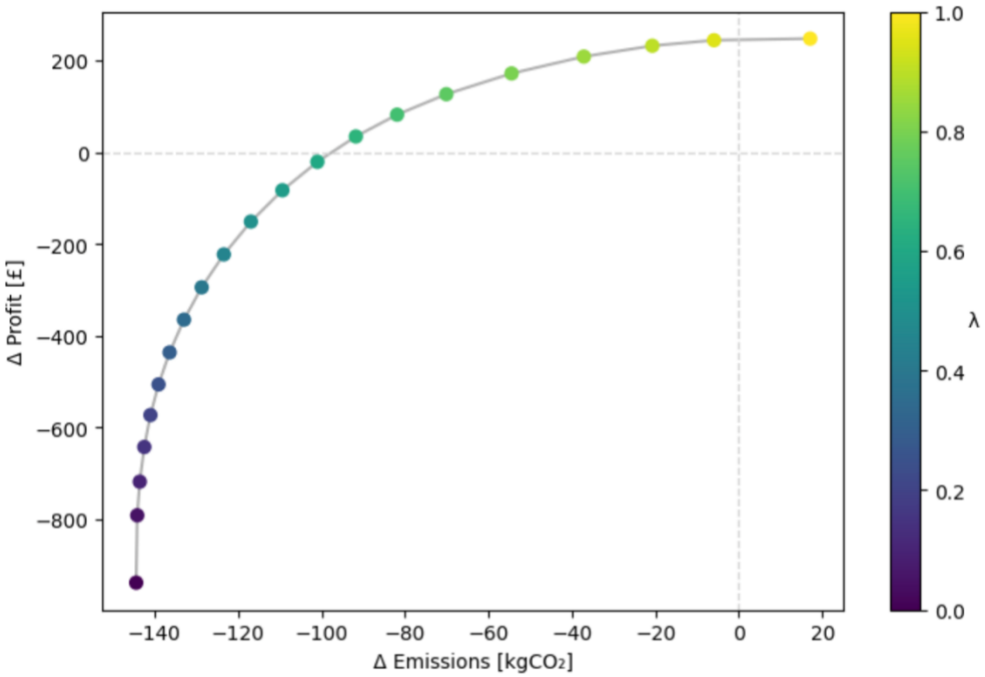

Figure 8: Changes in retail profits and CO_2 emissions following joint optimisation using various priority weightings, λ.

Given the previous findings, which suggest the existence of the merit order dilemma (Section 4.1), this is perhaps an unsurprising result. MEF estimates are lowest during peak demand when energy costs are highest. Increasing consumption in these periods will reduce CO_2 emissions but also reduces the available profit margin for the retailer when pricing caps are enforced.

However, whilst profit maximisation and emission minimisation might be opposing goals, Figure 8 illustrates the potential for the simultaneous improvement of both objectives relative to the baseline. On average, emissions can be reduced by up to 98kgCO_2 without reducing profits. In addition, retail profits can be raised by £245 without increasing CO_2 emissions. Therefore, in spite of a negative trade off, the joint optimisation pricing model provides a range of options which can satisfy both commercial and environmental motivations.

4.4Discussion

The outcomes of both price optimisation models are positive results regarding the application of PBDR to reduce CO_2 emissions. Not only does it reinforce findings from Pepiciello et al., (2019), it strengthens them, illustrating the feasibility of MEF-driven pricing models in the retail market.

However, in lieu of observable ground truth, the presented results are susceptible to the assumptions made in the modelling process. For example, the baseline state is modelled by a

uniform flat rate. If initial tariffs already incorporate price variation, optimised models may have less room for impact. But this works in both ways. By enforcing demand conservation, pricing strategies can only reduce emissions via load shifts. In reality, consumption curtailment during higher prices would likely yield further reductions.

A further simplification is the assumption of perfect foresight. The same MEF estimates are used to both set prices and to evaluate their impact. In practice, prices are set ahead of time. Inevitable differences between forecasted and realised MEFs will yield less precise price-setting, and lead to less certain CO_2 savings.

Due to insufficient price variation in the smart meter dataset, consumer demand functions were developed exogenously. The elasticity matrices used, model basic demand shifting and include limited variation between consumer groups. Incorporating more complexity into these structures presents a way this study can be enhanced moving forward.

To develop its results, the study considers a residential district of 3000 households. Load shifting from this population size is unlikely to affect contributions to the marginal mix. However, the applications of these pricing models at a larger scale would require dynamic modelling of MEFs in response to significant movements in aggregate demand.

5 Conclusion

This paper provides a comprehensive analysis of the potential for PBDR to reduce CO_2 emissions from the UK energy system. To achieve this, it begins by producing regression-based estimations of MEFs for the UK energy system. Alongside, exogenously modelled customer demand functions, estimates are then used for the optimisation of two multiple dynamic pricing models.

Regression outputs illustrate a significant temporal variation of MEFs with lowest values corresponding to times of peak energy demand. Subsequently, MEF-driven prices incentivise increased consumption during high system loads, ensuring that marginal consumption derives from cleaner energy sources which hold lower priority in the merit order.

An emission minimising price optimisation model achieves a 144kgCO_2 emission reduction, on average, relative to the baseline. However, this is achieved at the expense of 50% reduction in retail profits. When retail profits were held equivalent to the baseline, joint optimisation achieved a mean reduction of 98kgCO_2. Running joint optimisation using multiple weights illustrates a negative trade-off between the objectives but also highlights a significant scope for simultaneous improvements compared to the baseline.

Overall, the main contributions to the literature are twofold. First it supports the conclusions of Pepiciello et al. (2019) that MEF-driven pricing models can be a powerful tool for reducing CO_2 emissions from electricity consumption. Second, it demonstrates, to a significant extent, that these reductions are economically feasible when retail profits are considered within joint optimisation pricing strategy.

To develop the discussion further, future research might consider more complex modelling of consumer behaviour to create more realistic load shifts in response to price changes. In addition, repeating the methodology for other energy systems and increasing analysis to cover a larger population base would offer important tests of external validity.

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